1.

(a) $N=20$, $\Sigma y_i^2 = 7825.94$, $\bar{y} = 19.21$, $SSR = 375.47$

$$R^2 = \frac{SSR}{SSTO} = \frac{SSR}{\Sigma y_i^2 - N\bar{y}^2} = \frac{375.47}{7825.94 - 20\times 19.21^2} = 0.8429 \#$$

(b) $R^2 = 0.7911$, $SSTO = 725.94$, $N=20$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{(1-R^2)\times SSTO}{N-2} = \frac{0.2089\times 725.94}{18} = 8.4249 \#$$

(c) $\Sigma(y_i - \bar{y}_i)^2 = 631.63$, $\Sigma \hat{e}_i^2 = 182.85$

$$R^2 = \frac{SSE}{SSTO} = \frac{182.85}{631.63} = 0.7105 \#$$

3.

(a) $\hat{y} = 1.2 + 0.8x$

$x_0 = 4 \Rightarrow \hat{y}_0 = 1.2 + 0.8\times 4 = 4.4 \#$

(b) $$\widehat{Var}(f) = \hat{\sigma}^2\left[1 + \frac{1}{N} + \frac{(x_0-\bar{x})^2}{\Sigma(x_i-\bar{x})^2}\right] = 1.2\times\left(1 + \frac{1}{5} + \frac{(4-1)^2}{10}\right) = 2.52$$

其中 $\hat{\sigma}^2 = \frac{\Sigma \hat{e}_i^2}{n-2} = 1.2$, $\bar{x} = 1$

$\Rightarrow SE(f) = 2.52^{\frac{1}{2}} = 1.5875 \#$

(c) 95% P.I. for y give $x_0 = 4$: $\hat{y}_0 \pm t_{0.025(3)} \times SE(f)$

$\Rightarrow [4.4 - 3.182\times 1.5875,\ 4.4 + 3.182\times 1.5875]$

$= [-0.6514, 9.4514]$

(d) 99% P.I. for y give $x_0 = 4$: $\hat{y}_0 \pm t_{0.005(3)} \times SE(f)$

$\Rightarrow [4.4 - 5.841\times 1.5875,\ 4.4 + 5.841\times 1.5875]$

$= [-4.8726, 13.6726]$

(e) $\bar{x} = 1 \Rightarrow \hat{y} = 1.2 + 0.8\times 1 = 2$

$$SE(f) = \hat{\sigma}\sqrt{1 + \frac{1}{N} + \frac{(1-1)^2}{\Sigma(x_i-\bar{x})^2}} = 1.2$$

95% P.I. for y give $x = \bar{x} = 1$

$\Rightarrow [2 - 3.182\times 1.2,\ 2 + 3.182\times 1.2] = [-1.8184, 5.8184]$

預測點 x_0 離 $\bar{x}$ 愈遠，預測的不確定性愈高

∴ 区間越寬

25.

(a)

```
Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.32528 -0.19199  0.00122  0.21678  0.86609 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408 
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16

Log-Linear Slope: 9.019661 
Log-Linear Elasticity: 0.9833701 
```

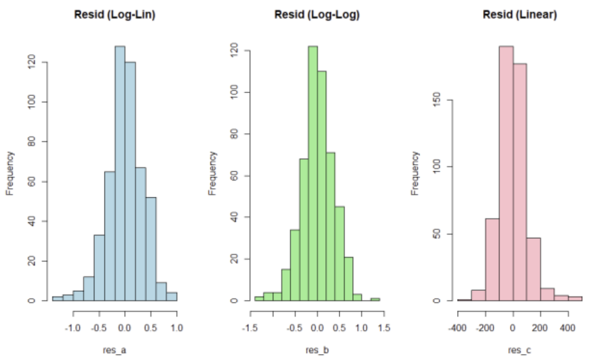
(b)

```
Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16

Log-Log Slope: 9.399923 
Log-Log Elasticity: 1.024828 
```

(c)

```
Linear R2: 0.6413167 
Log-Linear Generalized R2: 0.6621612 
Log-Log Generalized R2: 0.6445084 
```

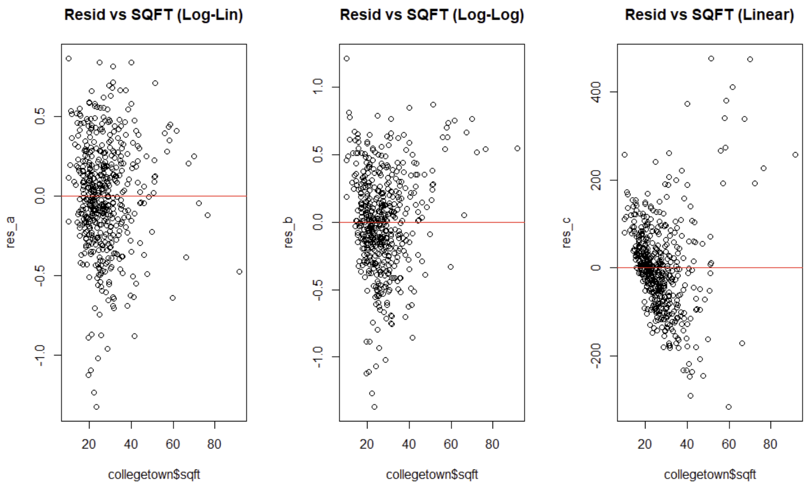
(d)




```
        Jarque Bera Test

data:  res_a
X-squared = 26.679, df = 2, p-value = 1.61e-06

        Jarque Bera Test

data:  res_b
X-squared = 14.279, df = 2, p-value = 0.0007933

        Jarque Bera Test

data:  res_c
X-squared = 221.11, df = 2, p-value < 2.2e-16
```

Although the histograms show that the residual distributions of the logarithmic models (Log-Linear and Log-Log) are closer to the normal than those of the linear models, the Jarque-Bera test results show that the p-values for all three models are well less than 0.05.

(e)



Observing the scatter plot of the residuals against the SQFT reveals that the Linear model exhibits significant heteroskedasticity. Its residuals show a marked funnel-shaped spread with increasing house area, exhibiting a systematic curvilinear pattern, indicating that the linear function cannot accurately capture the relationship between house price and area. In contrast, the Log-Linear and Log-Log models effectively improve this problem. In particular, the Log-Log model has the most uniform and random residual distribution, essentially eliminating the tendency of the variance to change with the variable, and best conforming to the econometric assumption of homoskedasticity.

(f) & (g)

```
$Log_Linear
       fit      lwr      upr
1 214.2336 109.7685 418.1167

$Log_Log
       fit      lwr      upr
1 227.5386 111.1406 465.8406

$Linear
       fit      lwr      upr
1 246.4557 44.27727 448.6341
```

(h)

Considering various statistical indicators and economic practices, this study concludes that the Log-Log model is the most appropriate functional form. The reasons are as follows: First, in residual analysis, the Log-Log model has the lowest Jarque-Bera statistic (14.279), indicating that its residuals are closest to the normal distribution and effectively addressing the severe heterogeneity problem in linear models. Second, although the generalized R-squared performance of each model is similar, the Log-Log model can capture the nonlinear characteristic of diminishing marginal utility of housing prices as area increases. Finally, the "elasticity" interpretation provided by this model is more consistent with the pricing logic of the real estate market than that of linear models.